

Department of Electronic and Telecommunication
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University of Moratuwa, Sri Lanka



Brain-Computer Interface for Locked-In Pediatric Patient

MAIN SUPERVISOR:	MEMBERS:	GROUP 21
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Final Year Project Monthly Report (Month 5: November)
submitted in partial fulfillment of the requirements for the course module
EN4203/BM4201

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1 Summary of Individual Work Carried Out, Problems Encountered and Solutions

1.1 Gammune D.J.T

During this month, my work focused on the continuation of signal processing and interface development activities. I contributed to the initial interface development for signal visualization, integrating digital filtering techniques to improve EEG signal clarity. The interface was tested using recorded signals from the Analog Front-End system to verify correct visualization of EEG waveforms after filtering. A challenge encountered was tuning filter parameters to balance noise reduction while preserving relevant EEG features. This was addressed by iterative testing with different filter configurations and visual inspection of outputs.

1.2 Kumarasinghe R.D

During this month, my work focused on both patient-related preparation and Analog Front-End (AFE) hardware integration. I contributed to integrating the AFE hardware with the custom-designed headset, ensuring proper electrode placement, mechanical stability. This included supporting system-level testing to verify compatibility between the headset, electrodes, and acquisition hardware. In parallel, I was involved in refining the headset design to ensure comfort, safety and on addressing ethics review comments, including clarifications on data retention and destruction, post-trial support, and the separation of clinical care and research roles. These revisions were incorporated into the participant information and consent documents.

A key issue encountered was managing noise introduced by electrode movement and cable handling. This was mitigated through mechanical stabilization of electrode mounts.

1.3 Weerasinghe C.N

During this month, my work primarily focused on electrical debugging, validation, and integration of the Analog Front-End (AFE) hardware. I evaluated the EEG acquisition PCBs received from the previous project group to assess their suitability for system integration. Through electrical testing and signal validation, it was identified that only 2 out of the 3 received AFE PCBs were fully functional, while the remaining PCB exhibited electrical faults and could not be reliably used.

I carried out detailed electrical checks, including power integrity verification, signal continuity testing, and noise assessment, to confirm the operational status of the working boards. Based on these results, the system configuration was adjusted to proceed with the two functional PCBs while ensuring stable signal acquisition.

1.4 Wijewickrama W.K.D.D

This month, I continued development of the Analog Front-End firmware. The firmware was updated to support stable data acquisition from multiple EEG channels and real-time data transmission to the processing interface. Testing was carried out

using the assembled hardware setup, and waveform outputs were verified through the visualization interface. Timing and synchronization issues observed during early testing were resolved by adjusting sampling configurations and communication routines.

2 Overall Project Monthly Update

During November, the project progressed from planning and mid-evaluation preparation toward practical system validation and deployment readiness. Significant effort was dedicated to integrating hardware, firmware, and software components into a coherent system suitable for patient testing.

The Analog Front-End hardware and firmware were successfully tested together, demonstrating stable waveform acquisition and real-time visualization. Initial digital filtering and interface development enabled clearer observation of signals, supporting future experiments such as alpha rhythm detection.

Ethics-related revisions were addressed following reviewer feedback, including updates on data retention policies, post-trial support, and transparency regarding confidentiality.

Additionally, the project received grant-related support, which will enable procurement of essential EEG acquisition hardware and active electrode systems.

Overall, the project is now well-positioned to move into controlled patient trials and advanced signal analysis in the upcoming phase.

Declaration

We declare that the information provided in this report is a true and accurate record of the work carried out during the stated month.

Name	Signature	Date
Gammune D.J.T		14/12/2025
Kumarasinghe R.D		14/12/2025
Weerasinghe C.N		14/12/2025
Wijewickrama W.K.D.D		14/12/2025

Supervisor's Comments and Signature

Comments:



Dr. Pranjeevan Kulasingham
Supervisor Signature

14/12/2025

Date